

PartA_S1_TestPlan.pdf

Test Plan – E-Commerce Web Application

Document Name: PartA_S1_TestPlan

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1. Scope

Testing will be performed on the following modules of the e-commerce web application:

- User Registration / Login / Logout
 - Product browsing, search and filters
 - Cart management (add/update/remove items)
 - Promo code application
 - Checkout and payment (Card, PayPal, COD)
 - Order confirmation and tracking
 - Admin dashboard (orders, products, reports)
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2. Objectives

The objectives of testing are:

- Verify that all key e-commerce workflows function correctly end-to-end.
 - Ensure calculations (cart total, tax, discount, promo codes) are accurate.
 - Validate payment and order placement flows without errors or duplicate orders.
 - Identify functional, UI, security, and performance issues before release.
 - Ensure compatibility across major browsers and devices.
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3. Out of Scope

The following items are not included in this test cycle:

- Internal validation of third-party payment gateway systems (bank/PayPal internal logic)
- Production monitoring and analytics validation

- PCI compliance certification testing
 - Hardware-level device testing beyond selected test devices
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4. Testing Types

The following testing types will be performed:

1. **Functional Testing** – Validate each feature as per requirement.
 2. **Regression Testing** – Ensure new changes do not break existing features.
 3. **Integration Testing** – Validate checkout/payment and admin reporting integrations.
 4. **Performance Testing** – Verify application stability under expected user load.
 5. **Security Testing** – Validate login security, SQL injection, session handling.
 6. **Compatibility Testing** – Verify functionality across supported browsers/devices.
 7. **UAT Support Testing** – Support stakeholder validation of key flows.
 8. **Accessibility Testing** – Basic validation for keyboard navigation and form labels.
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5. Entry Criteria

Testing will begin when:

- Application build is deployed successfully on staging environment.
 - Test environment is stable and accessible.
 - Requirements/user stories are available and finalized.
 - Test data (users, products, promo codes) is available.
 - Payment sandbox credentials are provided.
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6. Exit Criteria

Testing will be completed when:

- All planned test cases are executed successfully.
 - All Critical and High severity bugs are fixed and retested.
 - Regression testing is completed.
 - Test summary report is prepared.
 - QA sign-off is provided for release readiness.
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7. Risks and Mitigation

Risk	Description	Mitigation
R1	Payment gateway sandbox is unavailable	Use fallback COD testing and verify via logs/mock responses
R2	Requirements change during testing	Maintain updated test cases and confirm changes with stakeholders
R3	Staging environment downtime	Escalate early, plan testing slots, use backup environment if possible
R4	Promo code/discount calculation issues	Add strong boundary and rule-based test coverage
R5	Admin reports mismatch (revenue/order data)	Validate using backend/API/database data comparison

8. Test Environment Requirements

Operating Systems

- Windows 10/11
- macOS (optional)
- Android 12+
- iOS 16/17

Browsers

- Google Chrome (latest)
- Mozilla Firefox (latest)
- Microsoft Edge (latest)
- Safari (iOS Safari)

Devices

- Desktop/Laptop
- Android phone
- iPhone

Test Data Requirements

- Valid and invalid customer accounts

- Admin account access
- Products (in-stock/out-of-stock)
- Promo codes (valid/expired/invalid)
- Test shipping addresses
- Sandbox payment credentials (card/PayPal)